```
candidates_json = [
  {
    "candidate_id": "C001",
    "name": "Ayesha Khan",
    "skills": ["Python", "Machine Learning", "Data Analysis", "Pandas", "SQL"],
    "education": "BS Computer Science",
    "experience": ["Data Analyst Intern", "Machine Learning Research Assistant"]
  },
  {
    "candidate_id": "C002",
    "name": "Ali Raza",
    "skills": ["React", "Node.js", "Express", "JavaScript", "MongoDB"],
    "education": "BS Software Engineering",
    "experience": ["Frontend Developer Intern", "Full Stack Trainee"]
  },
  {
    "candidate_id": "C003",
    "name": "Sara Ahmed",
    "skills": ["UI/UX Design", "Figma", "Adobe XD", "Prototyping", "User Researcl
    "education": "BS Design and Multimedia",
    "experience": ["UI Designer Intern", "Graphic Designer"]
  },
  {
    "candidate_id": "C004",
    "name": "Hamza Tariq",
    "skills": ["C++", "CUDA", "OpenMP", "Parallel Programming"],
    "education": "BS Computer Engineering",
    "experience": ["High-Performance Computing Project", "Teaching Assistant"]
  }
]

recruiter_json = [
  {
    "job_id": "J101",
    "title": "Machine Learning Intern",
    "description": "Assist in developing predictive models and data pipelines us
    "required_skills": ["Python", "Machine Learning", "Data Science"]
  },
  {
    "job_id": "J102",
    "title": "Full Stack Developer",
    "description": "Work on React and Node.js-based applications to develop scal
    "required_skills": ["React", "Node.js", "JavaScript", "Express"]
  },
  {
    "job_id": "J103",
    "title": "UI/UX Designer",
    "description": "Create user-friendly designs and prototypes for mobile and w
    "required_skills": ["UI/UX Design", "Figma", "Prototyping"]
```

```
  },
  {
    "job_id": "J104",
    "title": "Parallel Systems Intern",
    "description": "Work on GPU-based optimization and CUDA kernel design for hi
    "required_skills": ["C++", "CUDA", "OpenMP"]
  }
]
```

```
from sentence_transformers import SentenceTransformer, util
import json
```

```
model = SentenceTransformer('all-mpnet-base-v2')
```

```
modules.json: 100%                                                349/349 [00:00<00:00, 26.6kB/
                                                                  s]
config_sentence_transformers.json: 100%                           116/116 [00:00<00:00, 8.47kB/
                                                                  s]
README.md:        11.6k/? [00:00<00:00, 985kB/s]
sentence_bert_config.json: 100%                                   53.0/53.0 [00:00<00:00, 4.90kB/
                                                                  s]
config.json: 100%                                                 571/571 [00:00<00:00, 47.3kB/s]
model.safetensors: 100%                                           438M/438M [00:06<00:00, 107MB/
                                                                  s]
tokenizer_config.json: 100%                                       363/363 [00:00<00:00, 31.4kB/
                                                                  s]
```

```
candidates = candidates_json
jobs_internships = recruiter_json

print(candidates[0])
print(jobs_internships[0])
```

```
{'candidate_id': 'C001', 'name': 'Ayesha Khan', 'skills': ['Python', 'Machine Lea
{'job_id': 'J101', 'title': 'Machine Learning Intern', 'description': 'Assist in
```

```
def textify_candidate(c):
    return " ".join(c["skills"]) + " " + c["education"] + " " + " ".join(c["expe

def textify_job(j):
    return j["title"] + " " + " ".join(j["required_skills"]) + " " + j["descript
```

```
for c in candidates:
    for j in jobs_internships:
        sim = util.cos_sim(
            model.encode(textify_candidate(c)),
            model.encode(textify_job(j))
        )
        sim_score = float(sim)

        if sim_score >= 0.85:
            label = 4
            desc = "Excellent match"
        elif sim_score >= 0.75:
            label = 3
            desc = "Good match"
        elif sim_score >= 0.6:
            label = 2
            desc = "Moderate match"
        elif sim_score >= 0.5:
            label = 1
            desc = "Low match"
        else:
            label = 0
            desc = "Poor match"

        print(f"{c['candidate_id']} - {j['job_id']} | similarity: {sim_score:.2f}
```

```
C001 - J101 | similarity: 0.74 | match_label: 2 (Moderate match)
C001 - J102 | similarity: 0.16 | match_label: 0 (Poor match)
C001 - J103 | similarity: 0.21 | match_label: 0 (Poor match)
C001 - J104 | similarity: 0.14 | match_label: 0 (Poor match)
C002 - J101 | similarity: 0.15 | match_label: 0 (Poor match)
C002 - J102 | similarity: 0.72 | match_label: 2 (Moderate match)
C002 - J103 | similarity: 0.21 | match_label: 0 (Poor match)
C002 - J104 | similarity: 0.08 | match_label: 0 (Poor match)
C003 - J101 | similarity: 0.27 | match_label: 0 (Poor match)
C003 - J102 | similarity: 0.31 | match_label: 0 (Poor match)
C003 - J103 | similarity: 0.82 | match_label: 3 (Good match)
C003 - J104 | similarity: 0.11 | match_label: 0 (Poor match)
C004 - J101 | similarity: 0.25 | match_label: 0 (Poor match)
C004 - J102 | similarity: 0.17 | match_label: 0 (Poor match)
C004 - J103 | similarity: 0.11 | match_label: 0 (Poor match)
C004 - J104 | similarity: 0.73 | match_label: 2 (Moderate match)
```

Start coding or generate with AI.